JENNIFER KAPHAMMER

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M.S. Computer Science student focused on software engineering, machine learning, audio technology, and interactive systems; experience spanning distributed production infrastructure, computer vision research, and creative audio computing.

EDUCATION

University of Colorado, Boulder *Boulder, CO*
Masters of Science in Computer Science *August 2026 - May 2028 (Expected)*
Thesis-Based Research Program; research interests in computational bioacoustics, HCI, and signal processing.

University of Colorado, Boulder *Boulder, CO*
Bachelor of Arts in Computer Science *December 2025*
Bachelor of Arts in Music Technology & Media Studies
Cum Laude; GPA: 3.73/4.0; Dean's List (2022, 2025)

WORK EXPERIENCE

Spectrum *Greenwood Village, CO*
Systems Engineer I *February 2026 - August 2026*

- Developed and supported backend services in **distributed Linux environments**, improving system reliability and performance
- Diagnosed production issues by analyzing **logs, APIs, and service interactions** across multiple services
- Designed and enhanced **monitoring/logging pipelines (Splunk)** to improve observability and reduce time-to-resolution
- Automated workflows using **Bash scripting** with **SQL**, improving operational efficiency
- Supported deployment and scalability of **containerized applications in Kubernetes (Rancher)** environments

Spectrum *Greenwood Village, CO*
Systems Engineer Intern *May 2025 - August 2025*

- Built core components of a **cross-data center failover system** enabling high availability and resilient traffic routing across multiple regions
- Built **RESTful APIs in Maven-based Java Spring Boot** integrating with Oracle DB, Kubernetes (Rancher), and global load-balancing systems.
- Collaborated on testing, debugging, and deployment across **4 environments** in an Agile workflow

Peleg Lab - Biofrontiers Institute *Boulder, CO*
Research Assistant *June 2024 - May 2025*

- Built **machine learning** and **computer vision** models (**PyTorch**, **U-Nets**, **YOLO11**) with Python to track and predict bee movement
- Developed **image processing pipelines in MATLAB** to analyze bee swarm structure and dynamics
- Processed large image-derived datasets, transforming raw data into structured formats for analysis
- Collaborated with researchers to refine computational models and validate analytical results

SELECTED PROJECTS

Interactive Game Audio System (In-Progress) C++, Unreal Engine 5, Wwise

- Building an interactive game-audio demo focused on responsive sound behavior, runtime parameter control, and event-driven audio systems.

Zebrafish Phenotype Prediction Python, pandas, NumPy, scikit-learn

- Created a ZFIN-based ML Pipeline with large-scale one-hot encoding, class-imbalance-aware logistic regression, K-means clustering, and PCA for phenotype/genotype analysis.

Freesound Generative Performance JavaScript, Node.js, Freesound API, Web Audio API

- Built a browser-based generative audio system using the Freesound API and Web Audio API to collect tagged samples and dynamically select sounds for live audiovisual performances.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, JavaScript, MATLAB
ML/Scientific: PyTorch, scikit-learn, computer vision, CNNs, U-Net, YOLO11, PCA, pandas, NumPy
Systems/Web: Linux, Spring Boot, REST APIs, Docker, Kubernetes, Rancher, Git, CI/CD, PostgreSQL, Oracle
Audio/Creative Tech: Web Audio API, REAPER, SuperCollider, librosa, Logic Pro, Pro Tools, interactive and game audio development